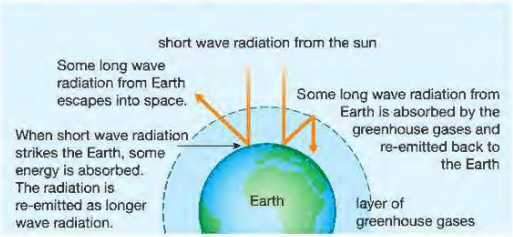
SKILLS REASONING

7 The diagram shows how the greenhouse effect is thought to operate.

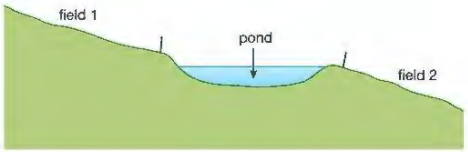


a Name two greenhouse gases.

b Explain one benefit to the Earth of the greenhouse effect.

c Suggest why global warming may lead to malaria becoming more common in Europe.

8 The diagram shows the profile of the ground on a farm either side of a pond.



The farmer applied nitrate fertiliser to the two fields in alternate years. When he applied the fertiliser to Field 1, the pond often developed an algal bloom. This did not happen when fertiliser was applied to Field 2.

a Explain why an algal bloom developed when he applied the fertiliser to Field 1.

b Explain why no algal bloom developed when he applies the fertiliser to Field 2.

c Suggest why the algal bloom is greater in hot weather.

9 Some untreated sewage was accidentally discharged into a small river. A short time afterwards, a number of dead fish are seen downstream of the point of discharge. Explain, as fully as you can, how the discharge might have led to the death of the fish.

10 Some farmers use pesticides and fertilisers to improve crop yields. Those practising 'organic' farming techniques do not use any artificial products.

a Describe how the use of pesticides and fertilisers can improve crop yields.

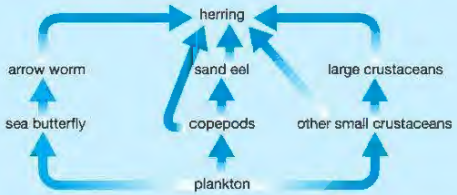
SKILLS CRITICAL THINKING

b Explain how organic farmers can maintain fertile soil and keep their crops free of pests.

UNIT QUESTIONS

SKILLS INTERPRETATION

1 The diagram shows a simplified food web of a fish, the herring.



a i Write out a food chain from the above food web containing four organisms. (1)

SKILLS ANALYSIS

ii From your food chain, name the primary consumer and secondary consumer. (2)

iii Name one organism in the web that is both a secondary consumer and a tertiary consumer. Explain your answer. (2)

b The amount of energy in each trophic level has been provided for the following food chain. The units are kJ per m^2 per year

plankton (8869) → copepod (892) → herring (91)

SKILLS INTERPRETATION

i Sketch a pyramid of energy for this food chain. (1)

SKILLS PROBLEM SOLVING

ii Calculate the percentage of energy entering the plankton that passes to the copepod. (2)

iii Calculate the percentage of energy entering the copepod that passes to the herring. (2)

iv Calculate the amount of energy that enters the food chain per year if the plankton use 0.1% of the available energy. (2)

SKILLS REASONING

v Explain two ways in which energy is lost in the transfer from the copepod to the herring. (2)

Total 14 marks

SKILLS INTERPRETATION (7)

2 A farmer added nitrate fertiliser to his wheat crop to try to increase yield. The table below shows the wheat yield when he added different amounts of nitrate fertiliser.

Fertiliser added / kg per hectare	Wheat yield / tonnes per hectare
0	15
50	18
100	22
150	31
200	30
250	31

a Use the information in the table to draw a line graph showing the effects of fertiliser on wheat yield. **(4)**

SKILLS PROBLEM SOLVING (6)

b What amount of fertiliser would you advise the farmer to use on his wheat crop? Explain your answer. **(3)**

SKILLS REASONING (5)

c Why is nitrate needed to help plants grow? **(1)**

(7)

d Excess nitrate can be washed into rivers. Explain the effects that this could have on the river ecosystem. **(5)**

Total 13 marks

SKILLS ANALYSIS, REASONING (7)

3 Insecticides are used by farmers to control the populations of insect pests. New insecticides are continually being developed.

a A new insecticide was trialled over three years to test its effectiveness in controlling an insect pest of potato plants. Three different concentrations of the insecticide were tested. Some results are shown in the table.

Concentration	Percentage of insect pest killed each year of insecticide		
	Year 1	Year 2	Year 3
1 (weakest)	95	72	18
2 (intermediate)	98	90	43
3 (strongest)	99	91	47

i Describe, and suggest an explanation for, the change in the effectiveness of the insecticide over the three years. **(3)**

(6)

ii Which concentration would a farmer be most likely to choose to apply to potato crops? Explain your answer. **(3)**

b The trials also showed that there was no significant bioaccumulation of the insecticide.

SKILLS CRITICAL THINKING (5)

i What is bioaccumulation? **(1)**

(6)

ii Give an example of bioaccumulation of an insecticide and describe its consequences. **(2)**

iii Explain why it is particularly important that there is no bioaccumulation of this insecticide. **(1)**

Total 10 marks

(8) **4** Carbon is passed through ecosystems by the actions of plants, animals and decomposers, in the carbon cycle. Humans influence the carbon cycle more than other animals.

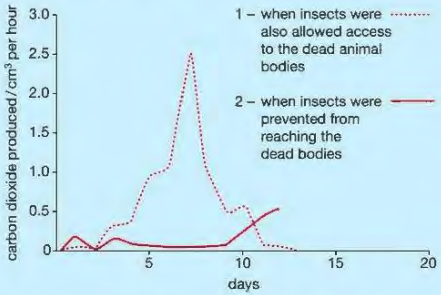
a Explain the importance of plants in the carbon cycle. **(2)**

(7)

b Describe two human activities that have significant effects on the world carbon cycle. **(2)**

SKILLS ANALYSIS, REASONING

c The graph shows the activity of decomposers acting on the bodies of dead animals under different conditions.



i Explain why carbon dioxide production was used as a measure of the activity of the decomposers. **(2)**

ii Describe and explain the changes in decomposer activity when insects were also allowed access to the dead bodies (curve 1) **(3)**

iii Describe two differences between curves (1) and (2). Suggest an explanation for the differences you describe. **(4)**

Total 13 marks

SKILLS REASONING (6) **5** In natural ecosystems, there is competition between members of the same species as well as between different species.

a Explain how competition between members of the same species helps to control population growth. **(3)**

b Crop plants must often compete with weeds for resources. Farmers often control weeds by spraying herbicides (weedkillers).

SKILLS CRITICAL THINKING (6)

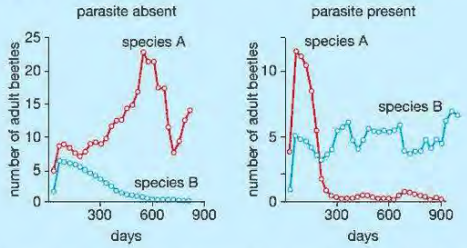
i Name two factors that the crop plants and weeds may compete for and explain the importance of each. **(4)**

SKILLS REASONING (7)

ii Farmers usually prefer to spray herbicides on weeds early in the growing season. Suggest why. **(2)**

SKILLS ANALYSIS 8

c Two species of the flour beetle, *Tribolium*, compete with each other for flour. Both are parasitised by a protozoan. The graphs show the changes in numbers of the two species over 900 days when the parasite is absent and when it is present.



i Which species is the most successful when the parasite is absent? Justify your answer. **(2)**

ii What is the effect of the parasite on the relative success of the two beetles? Suggest an explanation for your answer. **(4)**

Total 15 marks

SKILLS PROBLEM SOLVING 6 **6**

The table gives information about the pollutants produced in extracting aluminium from its ore (bauxite) and in recycling aluminium.

Pollutants	Amount / g per kg aluminium produced	
air	extraction from bauxite	recycling aluminium
sulfur dioxide	88 600	886
nitrogen oxides	139 000	6760
carbon monoxide	34 600	2440
Water		
dissolved solids	18 600	575
suspended solids	1600	175

a Calculate the percentage reduction in sulfur dioxide pollution by recycling aluminium. **(2)**

SKILLS REASONING 7

b Explain how extraction of aluminium from bauxite may contribute to the acidification of water hundreds of miles from the factory where the aluminium is extracted. **(3)**

c Suggest two reasons why there may be little plant life in water near an extraction. **(4)**

Total 9 marks

SKILLS INTERPRETATION 7 **7**

Rust fungi infect the leaves of many crop plants.

a Explain two ways in which an infection by rust fungi can reduce crop yield. **(2)**

b Infections of rust fungi can be controlled by spraying the crops with a fungicide. In an investigation into the effectiveness of this method of control, a field of wheat was divided into two plots. As soon as the first signs of infection appeared, one plot was sprayed with fungicide. The other plot was left unsprayed.

The relative amount of infection in the two plots was monitored over 50 days. The table shows the results of the investigation.

Day	Relative amount of infection / arbitrary units	
	Unsprayed (control)	Sprayed plants
0	0	3
10	17	8
20	32	10
30	48	15
40	62	17
50	65	18

SKILLS ANALYSIS 6

i Plot a line graph of these data. **(5)**

ii From your graph, find the relative infection of sprayed and unsprayed plants on day 24. **(2)**

SKILLS EXECUTIVE FUNCTION

iii Explain why it is necessary to have a control group. **(2)**

SKILLS ANALYSIS

iv The investigation was repeated every year for five years. Each year seeds were kept for planting the following year. On day 30 in Year 5, the relative amounts of infection were 46 (unsprayed) and 38 (sprayed). Compare these results with Year 1 and suggest an explanation for the difference. **(3)**

Total 14 marks